



Transaction-aware inverse reinforcement learning for trading in stock markets

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Abstract

Training automated trading agents is a long-standing topic that has been widely discussed in artificial intelligence for the quantitative finance. Reinforcement learning (RL) is designed to solve the sequential decision-making tasks, like the stock trading. The output of the RL is the policy which can be presented as the probability values of the possible actions based on a given state. The policy is optimized by the reward function. However, even if the profit is considered as the natural reward function, a trading agent equipped with an RL model has several serious problems. Specifically, **profit is only obtained after executing sell action, different profits exist at the same time step due to the varying-length transactions and the hold action deals with two opposite states, empty or nonempty position**. To alleviate these shortcomings, in this paper, we introduce a new trading action called *wait* for the empty position status and design the appropriate rewards to all actions. Based on the new action space and reward functions, a novel approach named Transaction-aware Inverse Reinforcement Learning (TAIRL) is proposed. TAIRL rewards all trading actions for avoiding the reward bias and dilemma. TAIRL is evaluated by backtesting on 12 stocks of US, UK and China stock markets, and compared against other state-of-art RL methods and moving average trading methods. The experimental results show that the agent of TAIRL achieves the state-of-art performance in profitability and anti-risk ability.

Keywords Finance · Transaction-aware · Inverse reinforcement learning · Algorithmic trading

1 Introduction

Quantitative finance [1] has been developed for almost 30 years. In Quantitative finance, financial analysts make use of the mathematical model and large data to analyze the financial market. One of the powerful mathematical models of the machine learning is the reinforcement learning [2] which is commonly used for solving sequential decision-making problems. These models are widely used in algorithmic trad-

ing [3–5]. The objective of reinforcement learning in stock trading application is to train an agent which automatically sends trading signals to the financial market based on the current situations of the financial market. The basic process of reinforcement learning is that the agent interacts with the environment and receives the reward from the environment to update the agent. This process is defined as Markov Decision Process (MDP) [2]. There are five elements in MDP: State (S), Action (A), Reward (R), Policy (π) and Discount Factor (γ). MDP is defined as (S, A, R, π, γ) . State refers to the observation of the environment. Action denotes the actions taken by the agent to interact with the environment. Reward is the feedback from the environment after an action is executed in the environment. Policy is the mapping from State to Action. Discount factor is used to discount the future reward into the current reward.

In machine learning, the workflow for solving a problem is to define an objective and optimize the designed algorithm to find the best parameters. As for reinforcement learning, the objective is to optimize the policy (π) which is the mapping from State to Action. However, reinforcement learning

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comes with the risk caused by the fact that the approximation function [2] may not accurately capture human intention. This leads to the agent behaving in an undesirable way. Usually the objective of investing in the stock market is to achieve a higher total value of the asset at the end of a trade. The total value is influenced by the Profit and Loss (PnL) of all transactions in the trading period. PnL is only obtained when a transaction is closed. For example, a transaction is initialized by a *buy* action and closed by a *sell* action. If PnL is only rewarded to the *sell* action in the training period, the *sell* action from an RL model will dominate the whole decision space for each time step. In this case, there will be no *buy* and *hold* actions except the *sell* action in the backtesting period. In addition, different profits exist at the same time step due to the varying-length transactions, and the *hold* action deals with two opposite states, empty or nonempty position, which increase the learning difficulty. The core of the rewarding bias is that all existing reinforcement learning algorithms [2] are developed based on the cumulative reward of (1), which cannot be easily replaced with other alternatives. By observing (1), it is very suitable to deal with the games. In the games [6–9] or robots [10–12], winning the game is rewarded 100 points no matter which action is taken at the end, the others are all 0. 100 points are discounted into the value function of every time step. However, finding an ultimate state like winning the game is difficult in the frequent stock trading. Besides, unlike the games, the training and testing datasets are different.

In order to alleviate these problems, in this paper, we propose a novel approach called Transaction-Aware Inverse Reinforcement Learning (TAIRL), which investigates the formation and elimination of transactions. Inverse Reinforcement Learning (IRL) [13] reverses MDP by executing the action first without considering the policy mapping from State to Action for obtaining the rewards during the training period. The reward of the stock trading is obviously related to profit, which is only obtained when a transaction is closed. But there are multiple profits at the same time step because a few transactions could be closed at same time step. These profits form an unknown distribution. According to the principle of the maximum entropy [14], when all probabilities are unknown, the probabilities of these profits are considered equal. To eliminate the influence of price, the return of a transaction is calculated by $\frac{\text{profit}}{\text{Buy Price}}$. Thereby, the reward is defined as the arithmetic mean of the returns at a given time step. Then, to avoid only rewarding the *sell* action, we assume that *buy* and *sell* have the same contribution to return. Taking the *hold* action in a transaction means the price trend is still upward in the future. But the risk of losing money increases along with the increase of the trading period. So we define the discounted return as the reward of the *hold* action. The discounted returns of the *hold* actions of different transaction also form an unknown distribution. So the

average discounted return is defined as the target value of the *hold* action. TAIRL takes both profitable and unprofitable transactions into consideration for avoiding the survivors bias which only considers the selected data in the previous researches [15, 16]. Besides, the uniqueness of price series leads to a fixed next state no matter which action is taken. In order to change this situation, the position status is introduced into the state. In this paper, we apply the multiple-input neural network to process two different types of the time series and the position status separately and newly add *wait* action against the commonly used actions (*buy*, *hold* and *sell*). The action *wait* takes over the empty position status. The action *hold* takes over the nonempty position status.

The trained agent of TAIRL is evaluated on the randomly selected 12 stocks from US, UK and Chinastock markets. In the experiments, TAIRL is compared with Deep Q Network (DQN) [17], Gated Deep Q Network (GDQN) [5], Proximal Policy Optimization (PPO) [18], Soft Actor Critic (SAC) algorithms [19], Simple Moving Average (SMA) [20] and Dual Moving Average Crossover (DMAC) [20]. GDQN substitutes the dense network of DQN with Gate Recurrent Unit (GRU) for processing the time series. PPO and SAC are policy-based reinforcement learning algorithms. Besides the reinforcement learning algorithms, the traditional trading policies SMA and DMAC are implemented as the benchmarks. The experiments on 12 stocks show that the trading results of the TAIRL agent are superior to the results of DQN, GDQN, PPO, SAC, SMA and DMAC in most cases. Experimental results show that the proposed novel trading strategy based on IRL achieves the state-of-art performance both in profitability and anti-risk ability. The contributions of this paper are summarized as follows.

1. In the context of Reinforcement Learning for trading in stock markets, the proposed approach considers a new action called *wait* in addition to other commonly used trading actions *buy*, *hold* and *sell*. The *wait* action is crucial in differentiating the empty and nonempty position status because it is taken before the *buy* action to take on the job that originally belongs to *hold* for the convenience of analyzing the learning ability and designing a new reward function. To the best of authors' knowledge, there is no existing work on distinguishing the empty and nonempty position in Reinforcement Learning for trading in stock markets.
2. In this paper, a novel approach named Transaction-Aware Inverse Reinforcement Learning (TAIRL) based on the new action space is proposed. TAIRL takes both profitable and unprofitable transactions into consideration for avoiding the survivors bias of the expert demonstration.
3. TAIRL achieves the state-of-art trading performance when compared to DQN, GDQN, PPO, SAC, SMA and

DMAC on randomly selected 12 stocks from the US, UK and China stock markets.

The rest of this paper is organized as follows. The Section 2 is the related works. The Transaction-Aware Inverse Reinforcement learning is detailed in Section 3. Section 4 is the experiments and performance of the proposed algorithm. The conclusion is made in Section 5.

2 Background and related works

Inverse Reinforcement Learning (IRL) is complementary to the Reinforcement Learning (RL) for resolving the problems of the objectives, values or the reward function. Transaction-Aware Inverse Reinforcement Learning (TAIRL) is proposed to devise a new reward function for avoiding the *sell* action bias during training the trading agent. In this section, RL, IRL and two traditional trading strategies used for comparison with the proposed method are briefly introduced.

2.1 Reinforcement learning

Reinforcement Learning [2] is similar to the human learning process in which the brain interacts with the environment continuously and learns the knowledge by being stimulated from the reward of the environment. Reinforcement learning is described by the Markov Decision Process (MDP) which is depicted in Fig. 1.

There are two subjects and three elements in Fig. 1. The two subjects are the Agent and the Environment. The agent is like a human and makes decisions based on the observation of the environment. The environment runs according to its own rules, which are unknown to the agent. State, Action and Reward are the three elements of MDP. State refers to the observation from the environment. Action is the decision made by the agent. Reward is defined as the feedback from the environment based on the given state and action. MDP is mathematically described as (S, A, R, π, γ) . π is the policy,

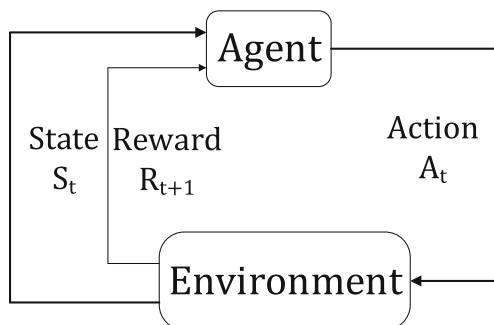


Fig. 1 Markov Decision Process (MDP)

which is the mapping from S to A . γ is the discounted factor. The process of Fig. 1 shows that the agent takes the action A_t based on the state S_t at time t and gets the reward R_{t+1} at the next time $t + 1$.

The objective of reinforcement learning is to find a policy that leads to the maximum cumulative rewards of (1).

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}, \quad (1)$$

where γ discounts the future reward as the part of the current reward. With the help of G_t , the state-action value function is defined as the expected value of the cumulative reward conditioned on the given state and action in (2). π is a policy.

$$Q_{\pi} \doteq \mathbb{E}[G_t | S_t = s, A_t = a] \quad (2)$$

The optimal policy (π^*) which leads to the optimal state-action value $Q^*(s, a)$ is given in (3).

$$Q^*(s, a) \doteq \max Q_{\pi}(s, a) \quad (3)$$

In 2015, Mnih et al. [7] proposed using the neural network as the policy that maps the state to the Q values and used the trained agent to defeat the world champion of Go. These value-based algorithms include the deep recurrent Q-Learning [17], dueling DQN [21], double DQN [22], rainbow DQN [9] and so on.

These reinforcement learning algorithms were evaluated on the games such as Atari games [23] and Go games [7], which only needs simple and hand-crafted reward functions. However, in most reinforcement learning tasks, there is no natural source for the reward function. To alleviate this problem, inverse reinforcement learning is proposed for solving this problem.

2.2 Inverse reinforcement learning for financial trading

Inverse reinforcement learning (IRL) [24–29] is designed to improve the objectives and rewards of the agent with the help of the action. The rewards of most existing researches [8, 9, 30, 31] are hand-crafted by the researchers. The goal of the inverse reinforcement learning is to offer a better reward function that is learned by imitating the expert behaviors [32]. The earliest imitation learning is behavior cloning, which records the trajectory of the human expert and repeats it for training the agent. However, behavior cloning simply imitates expert behaviors without reasoning about the causes behind these behaviors. IRL collects the behaviors of experts, reasons out the form of the reward function and finally optimizes the behavior policy by the reward function.

Ziebart et al. [24] proposed the maximum entropy inverse reinforcement learning to obtain the optimal policy by maximizing the formed entropy. The trajectory is defined in (4).

$$\tau = \{(s_1, a_1), \dots, (s_t, a_t), \dots, s_T\} \quad (4)$$

The reward function is defined in (5).

$$R_\theta = \sum_t r_\theta(s_t, a_t) \quad (5)$$

The expert demonstrations are defined as the collection $\mathcal{D} : \{\tau_i\} \sim \pi^*$. The maximum entropy formulation can be defined as follows:

$$p(\tau) = \frac{\exp(R_\theta(\tau))}{\int \exp(R_\theta(\tau)) d\tau} \quad (6)$$

The objective is to learn a optimal parameter θ which leads to the maximum value of the log probability in (7).

$$\max_{\theta} \sum_{\tau \in \mathcal{D}} \log p_{r_\theta}(\tau) \quad (7)$$

IRL is formulated as an optimization problem. However, the basic element of the optimization is the reward function of (5) which has different formulations for the different tasks. In the context of financial trading applications, Yand et al. [33] inferred the reward function for MDP from the observation of trading actions by using IRL. In addition, Linear IRL (LIRL) and Gaussian Process IRL (GPIRL) were deployed for categorizing and recognizing traders on the basis of observed limit orders. In 2018, Yand et al. [15] designed an investor sentiment reward-based trading system using Gaussian inverse reinforcement learning method for modeling the interaction mechanism between investor sentiment and market return. The market rewards of the investor sentiment are extracted by IRL and provide high quality signals to predict future market directions. In this paper, Transaction-Aware Inverse Reinforcement Learning (TAIRL) is proposed for defining the reward functions for all trading actions by averaging the returns when the return distribution is unknown. TAIRL is detailed in Section 3. In the following section, we briefly introduce two traditional trading strategies for comparison with the proposed method.

2.3 Baseline trading strategies

This paper proposes a novel trading strategy based on IRL. Two traditional trading strategies are selected as benchmarks for comparison: The Simple Moving Average (SMA) [34] and Dual Moving Average Crossover (DMAC) [34] trading strategies.

2.3.1 Simple moving average trading strategy

The Simple Moving Average (SMA) method [20] is a simple smoothing prediction technology that calculates the chronological average value of a certain number of items. SMA is defined as (8).

$$F_t = \frac{p_{t-n+1} + p_{t-n} + \dots + p_t}{n} \quad (8)$$

p is the price of the time series. F is the average of a certain period of prices. n is the number of moving average periods. The trading signals are initiated by the crossovers between the SMA and the price. The crossover is the line of SMA goes up or down through the line of the price. Assume that the price series is $[p_1, p_2, \dots, p_t, \dots]$. The *buy* signal is the up crossover of $F_{t-1} < p_{t-1}$ and $F_t > p_t$. The *sell* signal is the down crossover of $F_{t-1} > p_{t-1}$ and $F_t < p_t$.

2.3.2 Dual moving average crossover trading strategy

Dual Moving Average Crossover (DMAC) [20] utilizes two moving average lines with different periods to create the trading signals. The relative difference of the Short Term Moving Average (STMA) and the Long Term Moving Average (LTMA) is used to judge the buying and selling opportunities. When STMA pricks upward through LTMA in (9a), it is time for buying. Conversely, when STMA pricks downward through LTMA in (9b), it is time to sell.

$$\text{Trading Signal} = \begin{cases} \text{buy} , F_{t-1}^{STMA} < F_{t-1}^{LTMA} \& F_t^{STMA} > F_t^{LTMA} & (9a) \\ \text{sell} , F_{t-1}^{STMA} > F_{t-1}^{LTMA} \& F_t^{STMA} < F_t^{LTMA} & (9b) \end{cases}$$

SMA trading strategy and DMAC trading strategy are two frequently used moving average trading strategies. They are used as benchmarks for evaluating the proposed method named as Transaction-Aware Inverse Reinforcement (TAIRL) Learning, which is detailed in the following section.

3 Transaction-aware inverse reinforcement learning

In the context of Reinforcement Learning in games [6–9], the reward of winning the games is far higher than other rewards and they are hand-crafted. Therefore, these approaches are not suitable for financial trading because the objective of trading is to gain more profit. Besides, the agent will not be able to observe the situations learned from the training dataset from the testing dataset. Furthermore, the actions of reinforcement learning [2] are often considered to be independent. However, in the context of financial trading, the

buy action and the *sell* action indicate the beginning and the end of a transaction and they are not entirely independent. To alleviate these problems, a learning algorithm should retrospectively consider trading actions (previous transactions) for making new trading decisions. To this end, in this paper, we propose a novel Inverse Reinforcement Learning (IRL) approach which takes into account previous transactions for improving the reward functions.

According to MDP of Fig. 1, the trading agent of the reinforcement learning sends trading signals to the stock market based on current information. In response, the stock market returns information about the profit and next state to the agent. The stimulated market is driven by the historical financial market data. One unique characteristic of financial data is that it is impossible to reverse time to generate another time series. This leads to a constant next state in Fig. 1, regardless of the action taken. To have different states at each time step of the price time series, the state is defined as a combination of the time series (ϕ_{series}) and position status ($\phi_{position}$) as shown in (10).

$$S \doteq \{\phi_{series}, \phi_{position}\} \quad (10)$$

ϕ_{series} refers to the unique financial time series in this world, while $\phi_{position}$ represents the status of the position and can have a boolean value: either empty or non-empty.

3.1 State transitions

Due to the state definition in (10), there are two opposite states at each time step: the time series with an empty position and the time series with a position. The transition from the current state to the next state is triggered by the trading actions. In the proposed TAIRL, we define four trading actions. The process of the state transitions is illustrated in Fig. 2.

1. The *Wait* action triggers the transition from the current state with an empty position to the next state with an empty position. To the best of authors' knowledge, "Wait" trading action has not been considered in the existing work [3, 35, 36].
2. The *Buy* action involves purchasing the stock and transitioning from the current state with an empty position to the next state with a position.
3. The *Hold* action denotes doing nothing and maintaining the current state with a position to the next state with a position.
4. The *Sell* action involves clearing out the position and transitioning from the current state with a position to the next state with an empty position.

In Fig. 2, S_t^e represents the state S with an empty position at time t . The superscript e indicates an empty position, while n refers to a non-empty position. The subscript of S represents the time step. State transitions are driven by the trading actions shown by the arrows in Fig. 2.

3.2 Transaction-aware process

By following the state transition process, transactions are opened and closed by *buy* and *sell* actions. The agent makes trading decisions and takes the actions at every time step. However, the Profit and Loss (PnL) of a transaction is the collaborative result of the *buy* and *sell* actions. *PnL* is only obtained after selling the held stock. The *PnL* of a transaction is calculated as the difference between the sell price and the buy price, as shown in (11). *Cost* can be calculated based on the commission fee and the stamp duty.

$$PnL = Price_{sell} - Price_{buy} - Cost \quad (11)$$

The range of *PnL* is unlimited, and extremely high or low *PnL* values can lead to the gradient vanishing or explosion. To mitigate this issue, we propose using the return, as shown in (12) as the reward.

$$return = \frac{PnL}{Price_{buy}} \quad (12)$$

However, as shown in Fig. 3, there may be different transactions involving buying at different time steps and selling at the same time. These transactions can have different profits, leading to an unknown distribution at time step t .

3.3 Inverse reinforcement learning

As shown in Fig. 3, there is a reward distribution formed by the *returns* of these transactions at time t . The statistics of this distribution can be used to calculate the reward function using the inverse reinforcement learning (IRL). The learning process of IRL is opposite to the normal RL strategy of using reward function. Instead of using an existing reward function, IRL attempts to learn a reward function and use it to guide RL process. Traditionally, IRL follows the expert's trajectory to find the corresponding reward. Yang et al. [15] used IRL to extract market rewards based on investor sentiment and provided the sentiment rewards for predicting future market directions. Zhang et al. [16] applied IRL to calculate the expected value of the tick price returns over a predefined period as the reward for tick trading. However, the expert trajectory has the limited experience biased towards the successful cases and seldom considers the failed cases. In this paper, our expert demonstration considers all transactions, regardless of whether they are profitable or not. We

Fig. 2 The process of the state transitions

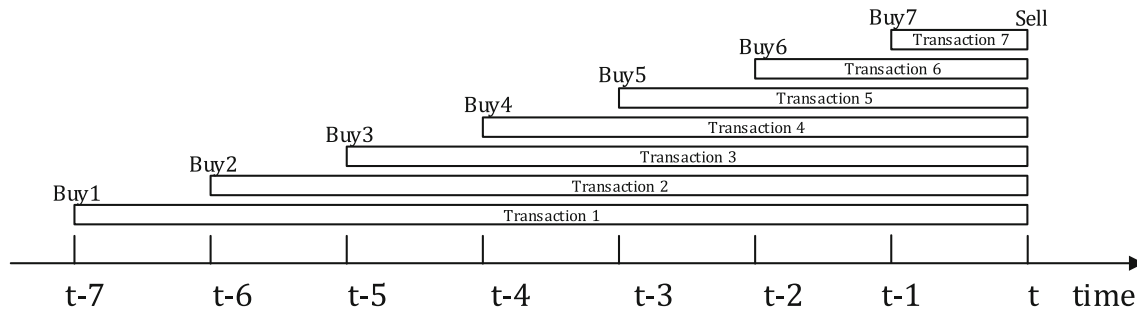
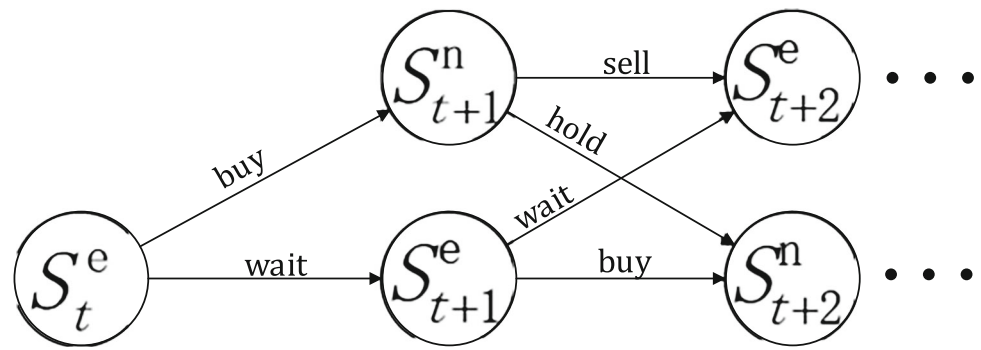


Fig. 3 Different transactions selling at the same time step t

first restrict the learning process within the life of the transactions. Then, we define the expectation of *return* distribution as the target value (see (13)), where $P(r_\phi(s, a))$ is the probability of the reward $r_\phi(s, a)$ and ϕ represents a transaction.

$$Y(s, a) = \mathbb{E}(R(s, a)) = \sum_{\phi} P(r_\phi(s, a)) * r_\phi(s, a) \quad (13)$$

However, $P(r_\phi(s, a))$ is unknown. According to the principle of the maximum entropy [37], the probability of $r(s, a)$ for each transaction is equal. Based on (13), the target value for a given state and the action *wait* defined as the average of the absolute returns in (14). d is the self-defined return threshold. Transactions with low or negative *returns* should not be considered into the rewards for *buy*, *hold* and *sell*. k is the number of the transactions which satisfies $\text{return} < d$.

$$y(s_t, \text{wait}) = \frac{1}{k} \sum_{i=1}^k |\text{return}_i^t|, \text{return}_i^t < d. \quad (14)$$

When *returns* of the potential transactions are greater than or equal to d , the transactions are created. The target value of a given state s and the action *buy* defined in (15). m is the number of the transactions that satisfies $\text{return} \geq d$.

$$y(s_t, \text{buy}) = \frac{1}{m} \sum_{i=1}^m \text{return}_i^t, \text{return}_i^t \geq d \quad (15)$$

For a transaction, taking the *hold* is due to an upward price trend. However, as the *sell* action approaches, the strength of

the *hold* action weakens. Therefore, we discount the return of the transaction from the time step of the *buy* action to the time step of the *sell* action. The target value of the *hold* is defined as the average of the discounted returns for all the transactions involving the *hold* action, as shown in (16). n is the number of the transactions at the time step t .

$$y(s_t, \text{hold}) = \frac{\gamma^{t-t_{\text{buy}}} \text{return} + \sum_{i=1}^n Q(s_t, \text{hold})}{n+1} \quad (16)$$

The target value for the given state s_t and the action *sell* is defined in (17). l is the number of the transactions which are initialized at the time t .

$$y(s_t, \text{sell}) = \frac{1}{l} \sum_{i=1}^l \text{return}_i^t \quad (17)$$

The target values of (14), (15), (16) and (17) all depend on the transactions, which lead to the reward distributions for each action at every time step. A transaction is initialized by a *buy* action and terminated at a *sell* action. The trading interval between a *buy* action and a *sell* action is filled with the *hold* actions at each time step. The maximum trading interval determines the number of the transactions in a given period. When the time step of the *sell* action is first deter-

mined, the transactions that generate the reward for this time step can be traversed backward from this time step. When the time steps of the *buy* action and sell action are determined, the forward visiting process is activated to compute the target value in (16). The pseudocode for TAIRL is provided in Algorithm 1.

Algorithm 1: The pseudo code of TAIRL.

Data : the length (ι) of the historical financial time series
Input : maximum interval of a transaction (mi), return threshold (d)

```

1 for  $t_{sell}$  from 0 to  $\iota$  do
2   for  $t_{buy}$  from  $t_{sell} - mi$  to  $t_{sell} - 1$  do
3      $return = \frac{Price_{sell} - Price_{buy}}{Price_{buy}}$ 
4     if  $return < d$  then
5        $t_{wait} = t_{buy}$ 
6       The target value for the wait action is
7        $y(s_{t_{wait}}, wait) = \frac{1}{k} \sum_{i=1}^k |return_i^{t_{wait}}|$ 
8       Update the Q value for the wait action:
9        $Q(s_{t_{wait}}, wait) \leftarrow Q(s_{t_{wait}}, wait) + \alpha(y(s_{t_{wait}}, wait) - Q(s_{t_{wait}}, wait))$ 
10    end if
11    else
12      The target value for the buy action is
13       $y(s_{t_{buy}}, buy) = \frac{1}{m} \sum_{i=1}^m return_i^{t_{buy}}$ 
14      Update the Q value for the buy action:
15       $Q(s_{t_{buy}}, buy) \leftarrow Q(s_{t_{buy}}, buy) + \alpha(y(s_{t_{buy}}, buy) - Q(s_{t_{buy}}, buy))$ 
16      The target value for the sell action is
17       $y(s_{t_{sell}}, sell) = \frac{1}{m} \sum_{i=1}^m return_i^{t_{sell}}$ 
18      Update the Q value for the sell action:
19       $Q(s_{t_{sell}}, sell) \leftarrow Q(s_{t_{sell}}, sell) + \alpha(y(s_{t_{sell}}, sell) - Q(s_{t_{sell}}, sell))$ 
20      for  $t_{hold}$  between  $t_{buy} + 1$  and  $t_{sell} - 1$  do
21        The target value for the hold action is
22         $y(s_t, hold) = \frac{\gamma^{t_{hold} - t_{buy}} return + \sum_{i=1}^n Q(s_i, hold)}{n+1}$ 
23        Update the Q value for the hold action:
24         $Q(s_{t_{hold}}, hold) \leftarrow Q(s_{t_{hold}}, hold) + \alpha(y(s_{t_{hold}}, hold) - Q(s_{t_{hold}}, hold))$ 
25      end for
26    end else
27  end
28 end

```

4 Experiments

In this section, we evaluate the effectiveness of the TAIRL trading agent. The trained TAIRL agent is tested on the stocks from the US, UK and China stock markets for the backtesting. At the same time, the output action of DQN, GDQN, PPO and SAC are restricted for avoiding the domination of the *sell* action. Additionally, two traditional trading strategies, Simple Moving Average (SMA) and Dual Moving Average Crossover (DMAC), are backtested as benchmarks.

4.1 The structure of the neural network

TAIRL is designed for training a trading agent that transforms the observations into trading actions. The policy leaned by TAIRL is implemented using a neural network. The main purpose of the neural network is to learn the position status and state transitions described in Section 3.1. Additionally, **the neural network can extract the features from the financial time series and combine them with the position status to generate the trading actions.** To accommodate different types of data, a two-input neural network is used. The structure of the two-input neural network is shown in Fig. 4.

The structure of the neural network consists of three components: the position status, the time series and the union. The position status component only contains the boolean value of 0 or 1. The time series processing component is a stacked Long Short Term Memory (LSTM) network [38]. The union use a dense network to process the merged result of the position status and the time series to output four Q values corresponding to the four actions. The parameters of the neural network are summarized in Table 1.

To avoid the domination of the *sell* action in the reinforcement learning algorithms like DQN and GDQN [5], the output of the trained agent is restricted by the position status as follows:

$$output = \begin{cases} \arg_a \max(Q(s, wait), Q(s, buy)) & , \text{Empty position} & (18a) \\ \arg_a \max(Q(s, hold), Q(s, sell)) & , \text{Non-empty position} & (18b) \end{cases}$$

4.2 Parameters setting

The parameters used by TAIRL for the model training and testing are listed in Table 2.

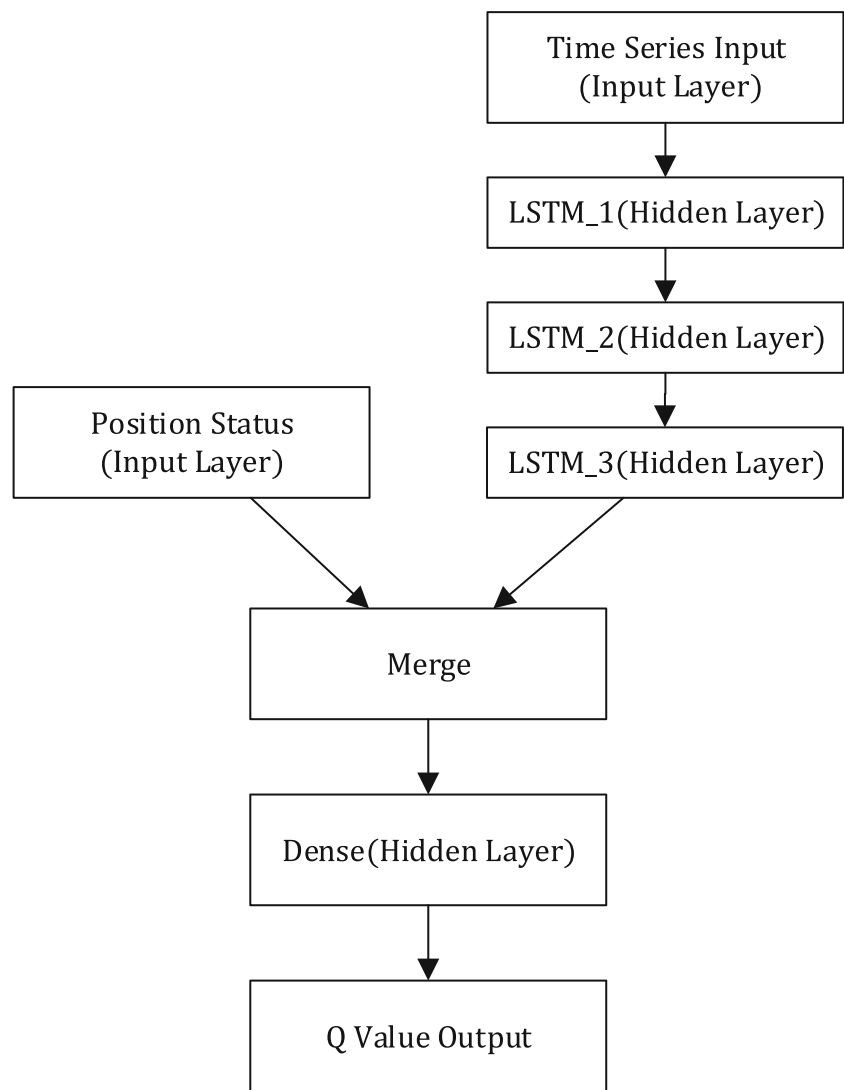
The discount factor (γ) is set by referring to existing reinforcement researches [4, 9, 39]. The transaction fee is set according to the quoted price of a US broker. To optimize the TAIRL model and achieve the best performance, the threshold (d) is tuned by using the grid search from -0.1 to 0.1. The maximum interval (mi) of a transaction is also tuned by the grid search from 5 days to 15 days. The best values of the threshold (d) and mi are found by grid search are 0 and 10 days, respectively.

After the TAIRL agent is trained based on the above settings, the agent is evaluated based on the following metrics.

4.3 Evaluation metrics

In this paper, we evaluate the TAIRL agent in terms of its profitability and the risk resistance ability. The commonly used metrics of the trading policy is listed as follows:

Fig. 4 The structure of the neural network for processing position status and time series data



1. **Total Value** is the sum of the asset value and the remaining cash.
2. **Annualized return** is the rate of return obtained over one-year investment.

$$\text{Annualized return} = \frac{\text{Investment reward}}{\text{Initial Investment}} * \frac{252}{\text{Investment days}} \quad (19)$$

Table 1 The parameters of the neural network

Layers	Output Size
Time Series Input	5
Position Status	3
LSTM_1	120
LSTM_2	60
LSTM_3	30
Dense	4

3. **Win rate** is the number of winning transaction (NWT) out of the number of total transactions (NTT).

$$\text{Win rate} = \frac{NWT}{NTT} \quad (20)$$

4. **The profit and loss ratio** is the rate of average profit divided by average loss. The number of loss transactions is abbreviated as NLT.

$$\text{profit/loss ratio} = \frac{\text{Total profit}}{NWT} \div \frac{\text{Total Loss}}{NLT} \quad (21)$$

5. **Sharpe Ratio** is defined as the difference between the returns of the investment (R_I) and the risk-free return (R_f), divided by the standard deviation (σ_I) of the invest-

Table 2 Parameters used by IRL for training and testing

Parameters	Discount factor(γ)	Transaction fee	Threshold (d)	mi
Value	0.95	0.003	0	10 days

ment returns.

$$\text{Sharpe Ratio} = \frac{R_I - R_f}{\sigma_I} \quad (22)$$

6. **Maximum DrawDown** refers to the maximum retracement range of the yield rate when the product net value reaches the lowest point after being pushed back.

4.4 Backtesting performance

This section presents the trading performance of the TAIRL agent when compared to DQN [17], GDQN [5], PPO [18], SAC [19], SMA [20] and DMAC [20]. The dataset used for training and testing were collected from the US, UK and China stock markets. The training dataset ranges from 2018-01-01 to 2020-12-31, while the test dataset ranges from 2021-01-01 to 2022-09-30. The initial investment for each stock is 100,000.

The trading results of the proposed method and the compared algorithms are analyzed both qualitatively and quantitatively. First, the *buy* and *sell* signals generated by TAIRL on the testing dataset are shown on the price series of each stock in the upper part of the figure, while the change in total values for all seven algorithms over the test period is shown in the lower part of the figure. Second, we provide detailed metrics for all seven algorithms at the end of the test period in tables. The datasets used in the experiments are categorized based on their corresponding stock markets, and the trading performance of the seven algorithms is presented in the following subsections.

4.4.1 US stocks

Four stocks are selected from US stock market and listed in Table 3. They are representatives of their respective industries.

SVT First, the results of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC trading on SVT are shown in Fig. 5. The

price series of SVT on Fig. 5(a) has three obvious segments. The first segment oscillates from Jan 2021 to Aug 2021, with a sudden price increase at the end. The second segment shows an upward oscillating price movement, followed by a plunge at the end. The third segment is more stable. The red triangles represent *buy* actions and the green inverted triangles represent *sell* actions on the price series.

The corresponding total value achieved by TAIRL is shown as a red line in Fig. 5(b), with a Maximum Drawdown of 19.75% annotated. Total values of DQN, GDQN, PPO, SAC, SMA and DMAC trading strategies are also shown in Fig. 5(b). From Fig. 5(b), we can observe that Total values of TAIRL is higher than that of DQN, GDQN, PPO, SAC, SMA and DMAC.

The performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC is listed in Table 4. TAIRL achieves the best values for Total Value, Annual Return, Sharpe Ratio, Max DrawDown and Profit/Loss ratio. A high Profit/Loss ratio and low win ratio indicate that TAIRL generates high profits from a small subset of positive transactions.

AAPL The price series of AAPL in Fig. 6(a) increase in the first half and decreases in the second half. On the AAPL price series, there exists a number of the *buy* and *sell* actions.

The performance metrics of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on AAPL are listed in Table 5. TAIRL achieves the highest values for Total Value, Annual Return, Sharpe Ratio and Win Rate.

FORD The price series of FORD shows a typical downward trend in Fig. 7(a). The total values earned by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC decline along with the decreasing price in Fig. 7(b). However, the total value of TAIRL is slightly higher than that of DQN, GDQN, PPO, SAC, SMA and DMAC.

The metrics for evaluating the trading performance of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC are listed in Table 6. From the experimental results, we can observe that Total value, Annual Return, Sharpe Ratio and

Table 3 Stocks from US stock market

Country	Ticker	Company	Industry
US	SVT	Servotronics, Inc.	Electrical Equipment
US	AAPL	Apple Inc.	Consumer Electronics
US	FORD	Forward Industries, Inc.	Footwear & Accessories
US	AMZN	Amazon.com, Inc.	Internet Retail

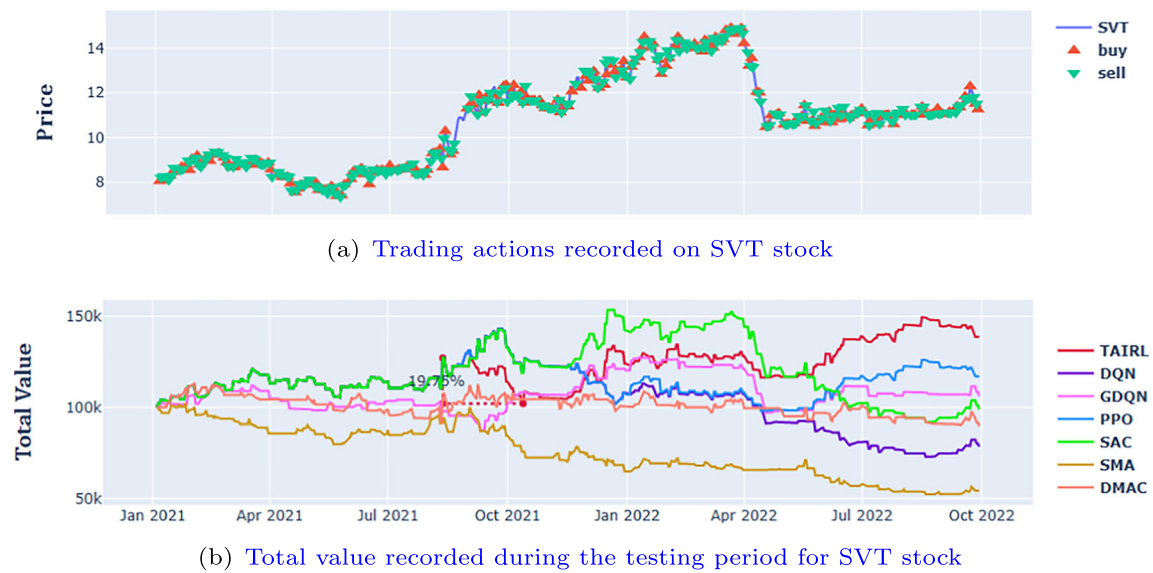


Fig. 5 The backtesting performance: Figure (a) shows the *buy* and *sell* actions taken by the TAIRL trading agent on the SVT price series, Figure (b) presents the total values of TAIRL, DQN, GDQN, PPO, SAC, SMA, DMAC

Table 4 Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on SVT

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	138,539.29	78,373.47	106,138.41	116,879.51	98,916.62	54,182.37	89,605.30
Annual Return	22%	-12%	4%	10%	-1%	-26%	-5%
Sharpe Ratio	0.68	-0.38	0.15	0.30	-0.02	-1.21	-0.19
Max DrawnDown	19.75%	49.15%	23.88%	31.43%	40.16%	49%	20%
Positive Trans	92	90	51	90	94	16	14
Negative Trans	122	127	66	129	119	48	21
Win Rate	42%	41%	43%	41%	44%	25%	40%
Profit/Loss ratio	1.59	1.26	1.36	1.56	1.26	1.43	1.18

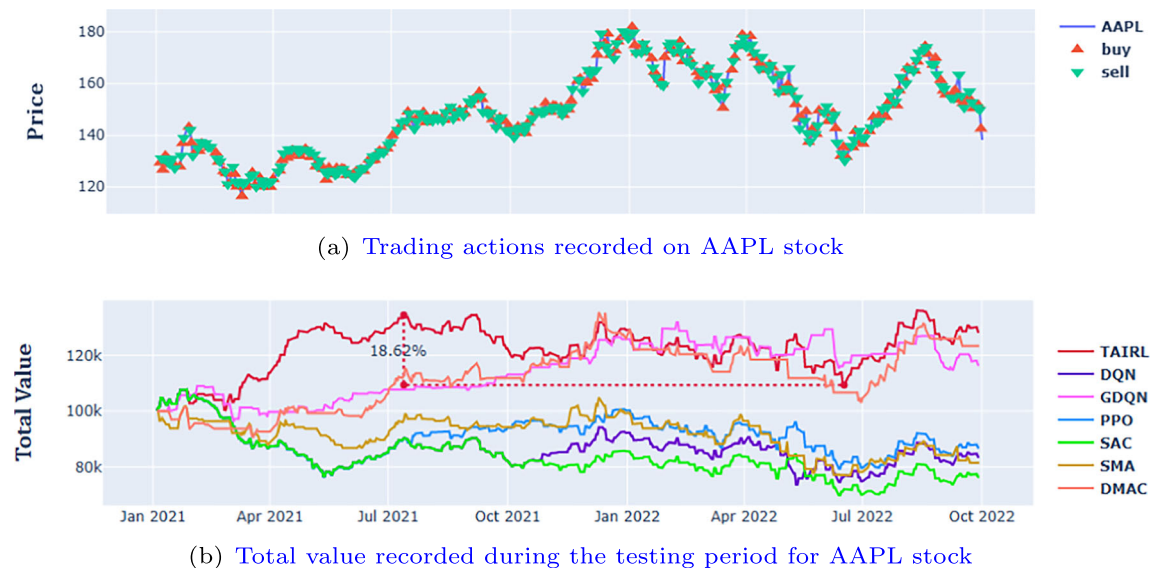
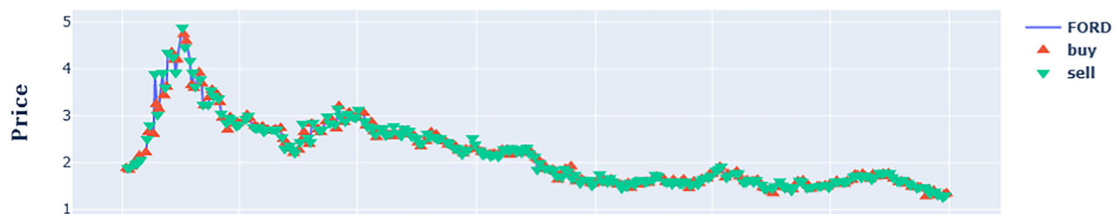


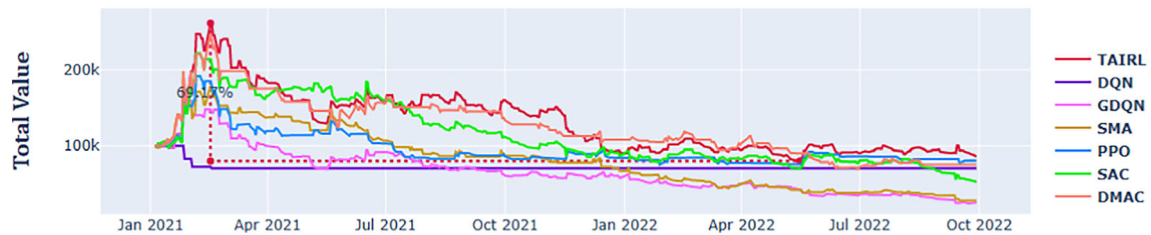
Fig. 6 Figure (a) shows the places of the *buy* and *sell* actions taken by TAIRL agent on the AAPL price series. Figure (b) shows the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC

Table 5 Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on AAPL

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	128,371.01	83,634.58	116,512.01	86,904.28	76,437.1	81404.22	123,242.17
Annual Return	16%	-9%	9%	-8%	-14%	-11%	13%
Sharpe Ratio	0.68	-0.52%	0.52	-0.42	-0.81	-0.59	0.63
Max DrawnDown	18.62%	31.62%	12.91%	28.96%	35.17%	26%	23%
Positive Trans	110	105	56	104	97	19	8
Negative Trans	106	114	36	107	117	42	16
Win Rate	51%	47.9%	60%	49.29%	45.33%	31%	33%
Profit/Loss ratio	1.13	0.96	0.78	0.93	1.00	1.67	3.11



(a) Trading actions recorded on FORD stock



(b) Total value recorded during the testing period for FORD stock

Fig. 7 Figure (a) shows *buy* and *sell* actions taken by TAIRL agent on the FORD price series. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC**Table 6** Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on FORD

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	86,646.59	71,135.39	26,410.47	81,029.59	53,316.48	28,166.79	75,726.34
Annual Return	-8%	-17%	-42%	-11%	-27%	-41%	-14%
Sharpe Ratio	-0.08	-1.13	-0.72	-0.17	-0.34	-0.53	-0.14
Max DrawnDown	69.17%	28.86%	85.67%	59.16%	75.94%	86%	72%
Positive Trans	103	1	93	29	92	14	6
Negative Trans	116	3	125	34	117	47	22
Win Rate	47.03%	25%	42%	46.03%	44.02%	22%	21%
Profit/Loss ratio	1.0987	0.007	1.02	1.06	1.12	1.89	3.19

Win Rate of TAIRL are all superior to those of DQN, GDQN, PPO, SAC, SMA and DMAC.

AMZN The price series of AMZN in Fig. 8 shows an increase in 2021 and a decrease starting from Jan 2022. The total value of the TAIRL agent on the AMZN price series is shown as a red line in Fig. 8(b).

The metrics for comparing the trading performance of TAIRL, DQN, GDQN, PPO, SAC, SMA, and DMAC are listed in Table 7. From the experimental results, we can observe that the Total Value, Annual Return, and Sharpe Ratio of TAIRL are the highest among all competitors. The lowest Max Drawdown of TAIRL indicates that its risk is the lowest among all models.

4.4.2 UK stocks

Information about four UK stocks is listed in Table 8. The first stock is BAE Systems, a weapons company in the aerospace and defense industry. The other three companies are all international, with prices forming different patterns. The trading performance of the TAIRL agent on these four stocks is detailed in the following paragraphs.

BAESY The price series of BAESY in Fig. 9(a) has many protuberances caused by special information and events. By observing the *buy* and *sell* actions taken by the TAIRL agent, it can be seen that the agent does not buy in advance and sell at the peaks of the protuberances. The total values in

Fig. 9(b) show that the TAIRL agent generates higher profits compared to DQN, GDQN, PPO, SAC, SMA, and DMAC.

The metrics in Table 9 provide more concrete evidence. From the experimental results, the Total Value, Annual Return, Sharpe Ratio, and Win Rate of TAIRL are the highest when compared to DQN, GDQN, PPO, SAC, SMA, and DMAC. The Max Drawdown of TAIRL is the lowest. The Win Rate of TAIRL reaches up to 65

IHG The price trend of IHG is shown to be downward in Fig. 10(a). The total values earned by the TAIRL agent are represented by the red line in Fig. 10(b), while the total values of DQN, GDQN, PPO, SAC, SMA, and DMAC are also shown. It is evident that the trading performance of the TAIRL agent is superior to that of DQN, GDQN, PPO, SAC, SMA, and DMAC at the end of the testing period.

In addition, the trading performance of TAIRL, DQN, GDQN, PPO, SAC, SMA, and DMAC is detailed in Table 10. According to the table, the Total Value, Annual Return, and Sharpe Ratio of TAIRL are higher than those of DQN, GDQN, PPO, SAC, SMA, and DMAC. The Max Drawdown of TAIRL is the lowest.

SMDS In Fig. 11(a), the price of SMDS is shown to increase before September 2021 and decrease afterward. In Fig. 11(b), the red line representing TAIRL increases as the SMDS price increases. However, the total values of DQN, GDQN, PPO, SAC, SMA, and DMAC decline with the decrease in price. The Max Drawdown of TAIRL is annotated between the peak and trough of the red line.



Fig. 8 Figure (a) shows the *buy* and *sell* actions taken by the agent of TAIRL on the AMZN price series. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC

Table 7 Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on AMZN

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	117,845.12	83,725.98	110,467.52	112,564.89	73,904.25	104,465.04	91,007.7
Annual Return	10%	-9%	6%	7%	-15%	2.5%	-5%
Sharpe Ratio	0.32	-0.39	0.26	0.28	-0.65	0.099	-0.21
Max DrawnDown	19.38%	44.75%	19.89%	31.99%	40.61%	30.30%	24.29%
Positive Trans	99	110	69	71	78	17	13
Negative Trans	75	109	52	49	68	37	12
Win Rate	56.89%	50%	57%	59.17%	53.42%	31%	52%
Profit/Loss ratio	0.8418	0.8965	0.8242	0.77	0.72	2.3	0.75

Table 8 Stocks from UK stock market

Country	Ticker	Company	Industry
UK	BAESY	BAE Systems	Aerospace & Defense
UK	IHG	InterContinental Hotels Group	Lodging
UK	SMDS	DS Smith	Packaging & Containers
UK	HSBA	HSBC Holdings	Banks



(a) Trading actions recorded on BAESY stock



(b) Total value recorded during the testing period for BAESY stock

Fig. 9 Figure (a) shows the *buy* and *sell* actions taken by the agent of TAIRL on the BAESY price series. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC**Table 9** Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on BAESY

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	134,310.88	110,952.13	100,606.74	102,554.76	98,697.52	104,374.57	95,692.83
Annual Return	20%	6%	0%	1%	-1%	2.5%	-2.4%
Sharpe Ratio	0.93	0.3	0.02	0.08	-0.05	0.12	-0.13
Max DrawnDown	13.97%	16.03	15.94%	19.03%	14.51%	19%	19%
Positive Trans	28	98	103	73	49	19	10
Negative Trans	15	74	74	56	45	41	16
Win Rate	65%	60%	58%	56.59%	52.13%	31%	34%
Profit/Loss ratio	0.999	0.8325	0.7227	0.79	0.9	2.34	1.43

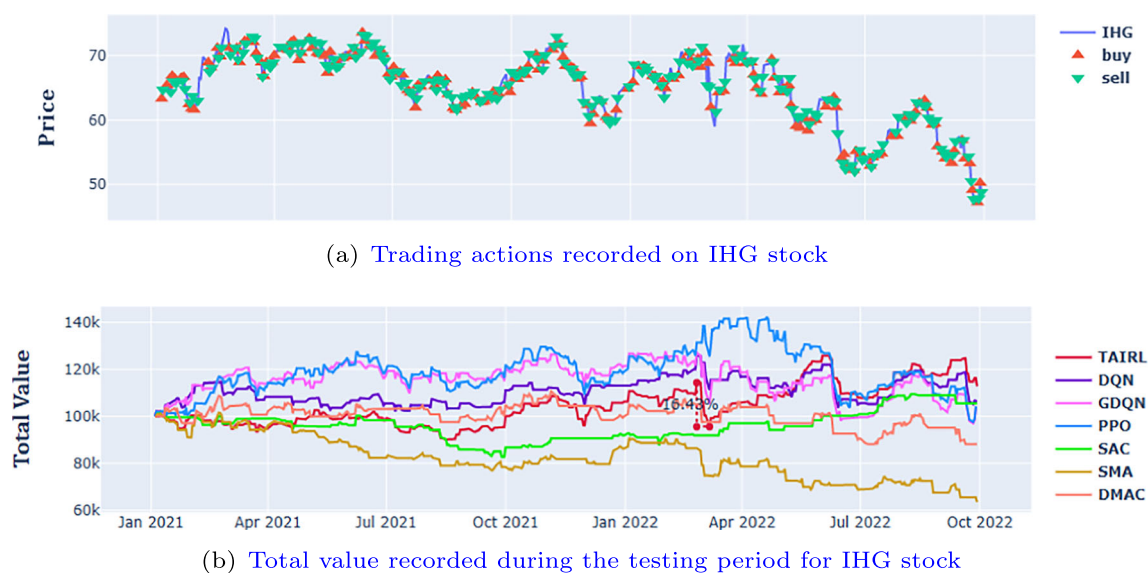


Fig. 10 Figure (a) shows the *buy* and *sell* actions taken by the agent of TAIRL on the IHG price series. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC

Table 10 Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on IHG

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	112,878.24	106,558.89	103,158.46	103,821.34	105,453.77	63,287.85	88,230.19
Annual Return	7%	4%	2%	2%	3%	-21%	-6%
Sharpe Ratio	0.33	0.16	0.06	0.07	0.26	-1.22	-0.32
Max DrawnDown	16.43%	17.43%	24.07%	31.46%	18.21%	37%	20%
Positive Trans	95	63	51	45	38	19	9
Negative Trans	91	36	32	37	33	41	13
Win Rate	51%	64%	61%	54.87%	53.52%	31%	42%
Profit/Loss ratio	1.0487	0.6095	0.6473	0.85	0.97	1.189	1.14

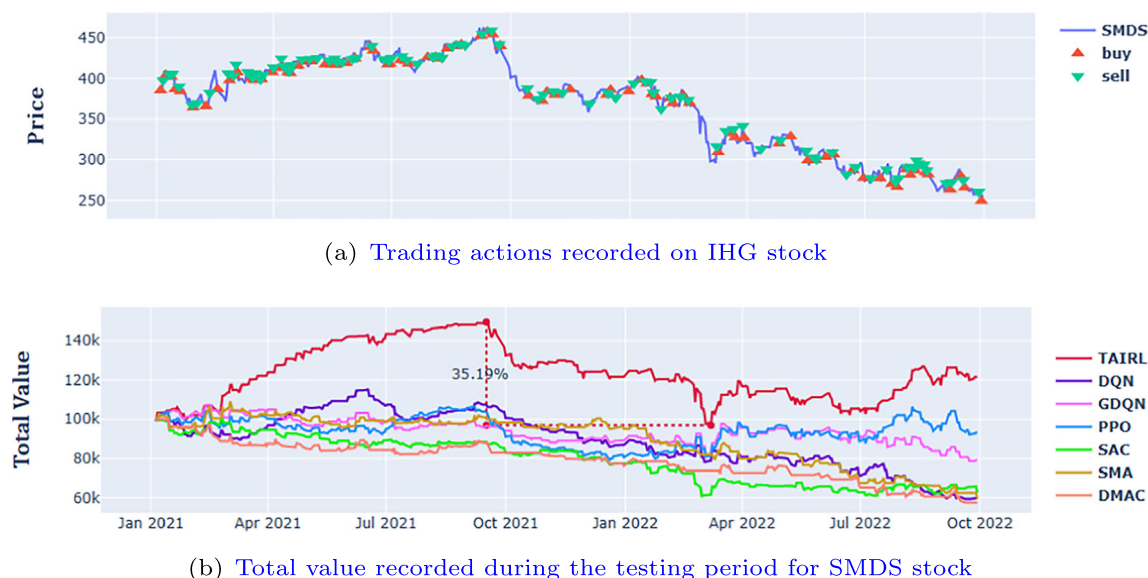


Fig. 11 Figure (a) shows the *buy* and *sell* actions taken by the agent of TAIRL on the SMDS price series. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC

The detailed performance of TAIRL, DQN, GDQN, PPO, SAC, SMA, and DMAC is presented in Table 11. According to the experimental results, the Total Value, Annual Return, Sharpe Ratio, and Win Rate of TAIRL are higher than those of DQN, GDQN, PPO, SAC, SMA, and DMAC.

HSBA In Fig. 12(a), the price of HSBA is shown to decrease initially and then increase. The number of *buy* and *sell* actions taken by the TAIRL agent is small. In Fig. 12(b), the red line representing TAIRL outperforms the other lines.

The trading performance of TAIRL, DQN, GDQN, PPO, SAC, SMA, and DMAC on HSBA is listed in Table 12. According to the experimental results, the Total Value, Annual Return, Sharpe Ratio, Win Rate, and Profit/Loss ratio of TAIRL are higher than those of DQN, GDQN, PPO, SAC, SMA, and DMAC.

4.4.3 China stocks

Four stocks from four different industries in the China stock market are selected and described in Table 13. The trading performance of the proposed TAIRL is detailed in the following sections.

660276 In Fig. 13(a), the price of 660276 is shown to decrease. The total values obtained by the seven algorithms are shown in Fig. 13(b).

The trading performance of TAIRL, DQN, GDQN, PPO, SAC, SMA, and DMAC is detailed in Table 14. According to the experimental results, the Total Value, Annual Return, and Sharpe Ratio of TAIRL are higher than those of DQN, GDQN, PPO, SAC, SMA, and DMAC.

600887 In Fig. 14(a), the price of 600887 is shown to decrease. The total values obtained by the seven algorithms are shown in Fig. 14(b).

The trading performance of TAIRL, DQN, GDQN, PPO, SAC, SMA, and DMAC is detailed in Table 15. According to the experimental results, the Total Value, Annual Return, and Sharpe Ratio of TAIRL are higher than those of DQN, GDQN, PPO, SAC, SMA, and DMAC.

600690 In Fig. 15(a), the price of 600690 is shown to decline with high oscillation. The total values of DQN, GDQN, PPO, SAC, SMA, and DMAC are shown in Fig. 15(b).

The trading performance among TAIRL, DQN, GDQN, PPO, SAC, SMA, and DMAC is detailed in Table 16. According to the experimental results, the Total Value, Annual Return, and Sharpe Ratio of TAIRL are the highest, and the Max Drawdown of TAIRL is the lowest when compared to DQN, GDQN, PPO, SAC, SMA, and DMAC.

600104 In Fig. 16(a), the price of 600104 is shown to decrease. The TAIRL agent takes only 11 *buy* actions and 10 *sell* actions on the price series. The total value of TAIRL, DQN, GDQN, PPO, SAC, and DMAC is shown in Fig. 16(b).

The trading performance of TAIRL, DQN, GDQN, PPO, SAC, SMA, and DMAC is detailed in Table 17. According to the experimental results, the Total Value of TAIRL is only lower than that of SMA and higher than that of DQN, GDQN, PPO, SAC, and DMAC. The Win Rate of TAIRL is the largest.

4.5 Discussion

In summary, we select 12 stocks for training a trading agent by TAIRL and compare the trading performance of TAIRL with the trading results of DQN [17], GDQN [5], PPO [18], SAC [19], SMA [20] and DMAC [20]. In all experiments, the *buy* and *sell* actions are annotated on the price series. The annotation of the *buy* and *sell* actions has the intuitive insight about the trading results. From Figs. 9, 11, 12 and 16, we

Table 11 Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on SMDS

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	121,439.95	59,872.56	79191.23	93,198.66	59,872.56	59,410.63	57,527.31
Annual Return	12%	-23%	-12%	-4%	-23%	-23%	-24%
Sharpe Ratio	0.44	-1.22	-0.61	-0.17	-1.22	-1.27	-1.62
Max DrawnDown	35.19%	48.6%	26.56%	25.67%	48.60%	45%	42%
Positive Trans	67	102	74	87	102	22	4
Negative Trans	20	117	88	80	117	41	21
Win Rate	77%	46.7%	46.6%	52.10%	46.57%	34%	16%
Profit/Loss ratio	0.3658	0.8572	0.9982	0.87	0.85	0.82	0.64

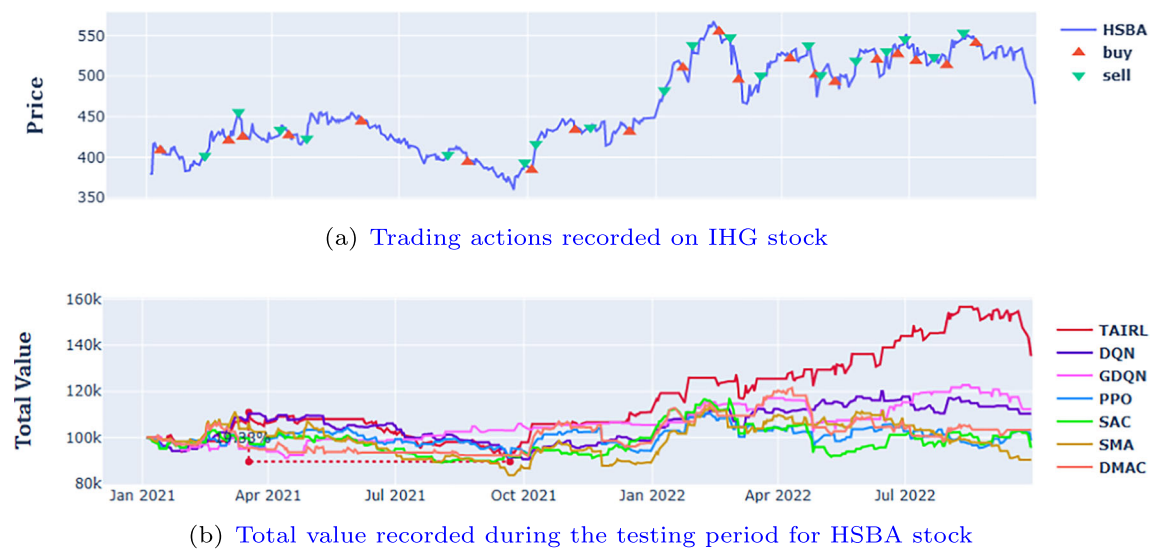


Fig. 12 Figure (a) shows the *buy* and *sell* actions taken by the agent of TAIRL on the HSBA price series. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC

Table 12 Performance comparison of TAIRL, DQN, GDQN, SMA and DMAC on HSBA

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	135,369.13	110,322.68	112,242.17	98,557.37	87,575.14	90,275.02	103,314.83
Annual Return	20%	6%	7%	-1%	7%	-5%	1.9%
Sharpe Ratio	0.82	0.38	0.44	-0.05		-0.29	0.1
Max DrawnDown	19.33%	17.83%	12.66%	13.72%	19.74%	24%	19%
Positive Trans	13	74	48	88	81	12	8
Negative Trans	6	78	33	80	69	49	16
Win Rate	68%	49%	59%	52.38%	54%	19%	33%
Profit/Loss ratio	2.28	1.1788	0.8433	0.89	0.79	3.56	2.15

Table 13 Stocks from China stock market

Country	Ticker	Company	Industry
China	600276	DJiangsu Hengrui Medicine Co., Ltd.	Drug Manufacturers
China	600887	Inner Mongolia Yili Industrial Group Co., Ltd.	Packaged Foods
China	600690	Haier Smart Home Co., Ltd.	Furnishings, Fixtures
China	600104	SAIC Motor Corporation Ltd	Auto Manufacturers

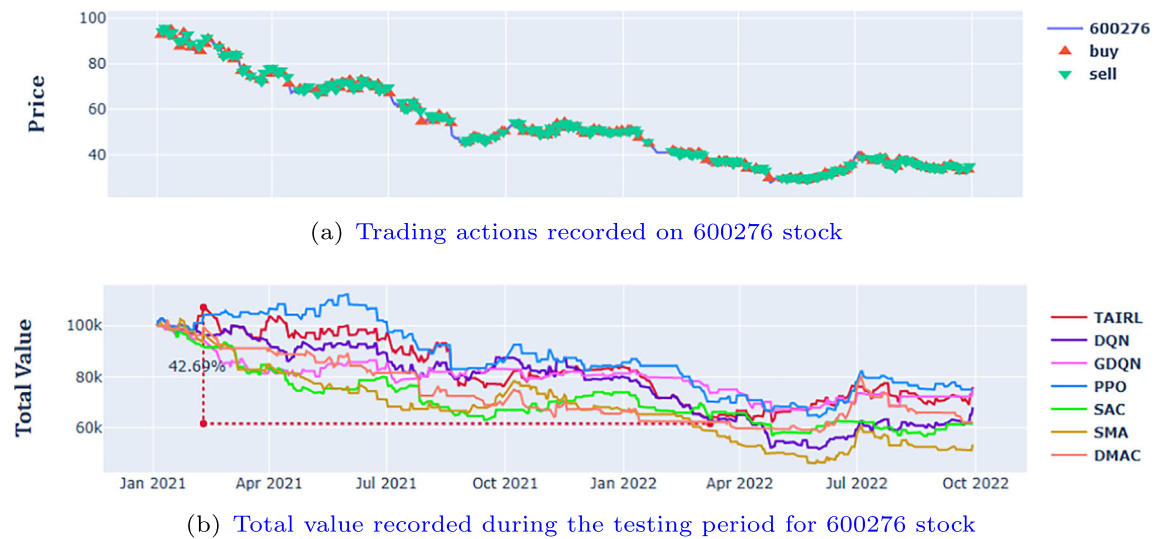


Fig. 13 Figure (a) shows *buy* and *sell* actions taken by TAIRL agent on the price series of 600276. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC

Table 14 Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on 600276

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	75,989.66	67,968.76	75,473.93	75,096.24	61,447.58	53,577.19	61,983.22
Annual Return	-14%	-19%	-15%	-15%	-23%	-27%	-22%
Sharpe Ratio	-0.62	-1.04	-0.99	-0.77	-1.59	-1.7	-1.39
Max DrawnDown	42.69%	49.83%	35%	42.96%	43.35%	55%	43%
Positive Trans	81	92	46	53	59	13	2
Negative Trans	95	120	51	65	77	46	20
Win Rate	46%	43%	47%	44.91%	43.38%	22%	9%
Profit/Loss ratio	0.98	1.23	0.7574	0.95	0.79	2.57	1.0202

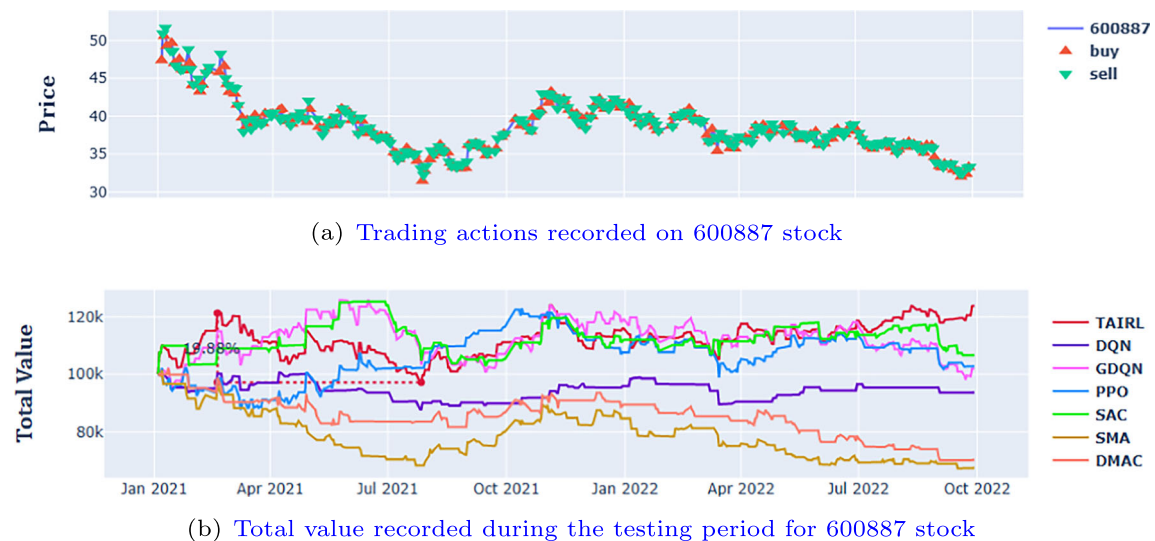
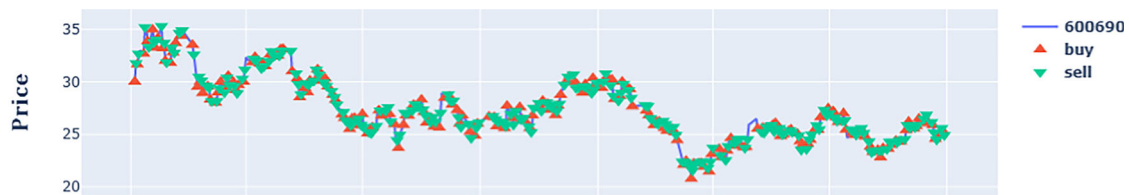


Fig. 14 Figure (a) shows *buy* and *sell* actions taken by the agent of TAIRL on the price series of 600887. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC

Table 15 Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on 600887

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	123903.22	93,672.18	101,910.10	102,840.82	106,649.89	66830.98	69,545.92
Annual Return	14%	-4%	1%	2%	4%	-19%	-18%
Sharpe Ratio	0.59	-0.33	0.04	0.08	0.19	-1.22	-1.11
Max DrawnDown	19.88%	12.68%	22.17%	19.61%	17.92%	34%	31%
Positive Trans	110	38	23	48	33	14	9
Negative Trans	102	31	17	37	17	41	18
Win Rate	52%	55%	46%	56%	66%	25%	33%
Profit/Loss ratio	1.07	0.7079	0.7589	0.79	0.57	1.56	0.64



(a) Trading actions recorded on 600690 stock



(b) Total value recorded during the testing period for 600690 stock

Fig. 15 Figure (a) shows the the *buy* and *sell* actions taken by the TAIRL agent on the price series of 600690. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN,PPO, SAC, SMA and DMAC**Table 16** Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on 600690

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	123,787.01	118,860.75	95,004.03	96,156.38	105,319.83	92,119.68	49,698.77
Annual Return	14%	11%	-3%	-2%	3%	4.6%	-29%
Sharpe Ratio	0.48	0.44	-0.1	-0.09	0.11	-0.19	-1.67
Max DrawnDown	21.24%	27.77%	31.74%	26.78%	36.06%	24%	56%
Positive Trans	110	107	33	50	31	22	6
Negative Trans	95	105	23	37	11	34	20
Win Rate	53%	50%	59%	57%	73%	39%	23%
Profit/Loss ratio	0.96	1.0818	0.6806	0.73	0.40	1.41	0.57



Fig. 16 Figure (a) shows the *buy* and *sell* actions taken by the agent of TAIRL on the price series of 600104. Figure (b) shows the comparison performance of the total values achieved by TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC

can clearly observe that the TAIRL agent buys or sells at the expected time steps. In order to provide the quantitative analysis of the trading results, 8 metrics are used for evaluating the trained agent of TAIRL. The profitability of TAIRL is evaluated by Total value, Annual Return, Win Rate and Profit/Loss ratio. The trading risk is evaluated by Max DrawnDown. Both the profitability and the risk are evaluated by Sharpe Ratio. Besides, in order to compare TAIRL with the traditional reinforcement learning algorithms, DQN, GDQN, PPO, SAC are implemented for the stock trading. Besides, two commonly used trading strategies in the financial industry, that are SMA and DMAC, are implemented as the benchmarks. From the experimental results, we can observe that TAIRL is superior to DQN, GDQN, PPO, SAC, SMA and DMAC in profitability and anti-risk ability in the single stock trading from the above 12 experiments.

5 Conclusion

Abbeel and Ng [40] suggested that the reward function of a control task must be succinct, robust for the reinforcement learning. By using appropriate reward functions, reinforcement learning model has achieved outstanding performance in the games, such as Atari games [23] and Go game [7]. The basic evaluation value of reinforcement learning is Q value. Q value is the expected cumulative reward function. The cumulative reward function is suitable to the tasks with a clear and quantitative objective like winning the games. High reward value set for winning guides the agent to make the right decision at each time step. Besides, the training environment and the testing environment are same in games [7, 23].

The reward in stock trading is often represented by the profit or its derivatives, which only rewards the *sell* action

Table 17 Performance comparison of TAIRL, DQN, GDQN, PPO, SAC, SMA and DMAC on 600104

Metrics	TAIRL	DQN	GDQN	PPO	SAC	SMA	DMAC
Total Value	81,651.7	68,706.42	74,621.43	80,395.96	67,829.64	85,984.89	52,226.66
Annual Return	-11%	-19%	-15%	-12%	-19%	-8%	-28%
Sharpe Ratio	-0.43	-0.77	-0.93	-0.51	-1.23	-0.41	-2.05
Max DrawnDown	33.22%	38.27%	30.29%	28.07%	33.67%	28%	48%
Positive Trans	6	46	54	75	9	18	5
Negative Trans	4	39	68	76	15	34	20
Win Rate	60%	54%	44%	50%	37.5%	34%	20%
Profit/Loss ratio	0.65	0.6536	0.8864	0.86	0.70	1.39	0.76

when a transaction is closed. This reward bias towards selling can lead to the domination of the *sell* action. In addition, different profits exist at the same time step due to varying-length transactions. The *hold* action deals with two opposite states: empty or nonempty position, which increases the learning difficulty. To alleviate these problems, we introduce a new trading action called *wait* to consider the state transition from the empty position to the next empty position while aiming to reduce the complexity of the *hold* action. Note that the *hold* action is used for the state transition from nonempty to next nonempty position. Besides, we extract the return distribution of related transactions and use the algorithmic mean of the return distribution as the reward for updating the Q value of the trading actions. Based on the new trading action space and the new reward function and target value function, a novel approach called Transaction-Aware Inverse Reinforcement Learning (TAIRL) is proposed in this paper.

The proposed TAIRL is backtested on 12 stocks of different industries from US, UK, and China stock markets. The trading process of TAIRL is analyzed qualitatively in the experiments. The trading performance is also evaluated quantitatively by 8 metrics on profitability and anti-risk ability. The trading results and performance of TAIRL are compared with DQN, GDQN, PPO, SAC, SMA, and DMAC. Experimental results show that TAIRL outperforms DQN, GDQN, PPO, SAC, SMA, and DMAC and achieves state-of-the-art trading performance in single stock trading.

Although the TAIRL agent has achieved outstanding results, the number of total transactions and the number of negative transactions are still high. Reducing the transaction frequency and improving the Win rate and profit/loss ratio are future work.

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References

- Mariani MC, Florescu I (2019) Quantitative Finance. John Wiley & Sons, London
- Sutton RS (2020) Sutton & Barto book: reinforcement learning: an introduction. In: A Bradford Book. MIT Press Cambridge, MA, London
- Liu X-Y, Yang H, Gao J, Wang CD (2021) Finrl: deep reinforcement learning framework to automate trading in quantitative finance. In: Proceedings of the second ACM international conference on AI in finance, pp 1–9
- Li Z, Liu X-Y, Zheng J, Wang Z, Walid A, Guo J (2021) Finrl-podracr: high performance and scalable deep reinforcement learning for quantitative finance. In: Proceedings of the second ACM international conference on AI in finance, pp 1–9
- Wu X, Chen H, Wang J, Troiano L, Loia V, Fujita H (2020) Adaptive stock trading strategies with deep reinforcement learning methods. Information Sciences 538:142–158. <https://doi.org/10.1016/j.ins.2020.05.066>
- Bellemare MG, Naddaf Y, Veness J, Bowling M (2013) The arcade learning environment: An evaluation platform for general agents. Journal of Artificial Intelligence Research 47:253–279
- Mnih V, Kavukcuoglu K, Silver D, Rusu AA, Veness J, Bellemare MG, Graves A, Riedmiller M, Fidjeland AK, Ostrovski G et al (2015) Human-level control through deep reinforcement learning. Nature 518(7540):529–533
- Chou P-W, Maturana D, Scherer S (2017) Improving stochastic policy gradients in continuous control with deep reinforcement learning using the beta distribution. In: International conference on machine learning, PMLR pp 834–843
- Hessel M, Modayil J, Van Hasselt H, Schaul T, Ostrovski G, Dabney W, Horgan D, Piot B, Azar M, Silver D (2018) Rainbow: combining improvements in deep reinforcement learning. In: Thirty-second AAAI conference on artificial intelligence
- Niroui F, Zhang K, Kashino Z, Nejat G (2019) Deep reinforcement learning robot for search and rescue applications: Exploration in unknown cluttered environments. IEEE Robotics and Automation Letters 4(2):610–617
- Marchesini E, Farinelli A (2022) Enhancing deep reinforcement learning approaches for multi-robot navigation via single-robot evolutionary policy search. In: 2022 international conference on robotics and automation (ICRA), IEEE pp 5525–5531
- Nguyen H, La H (2019) Review of deep reinforcement learning for robot manipulation. In: 2019 third IEEE international conference on robotic computing (IRC), pp 590–595. <https://doi.org/10.1109/IRC.2019.00120>
- Ng AY, Russell S et al (2000) Algorithms for inverse reinforcement learning. In: ICML, vol 1, p 2
- Alpaydin E (2020) Introduction to Machine Learning. MIT press, US
- Yang SY, Yu Y, Almahdi S (2018) An investor sentiment reward-based trading system using gaussian inverse reinforcement learning algorithm. Expert Systems with Applications 114:388–401
- Zhang W, Zhang N, Yan J, Li G, Yang X (2022) Auto uning of price prediction models for high-frequency trading via reinforcement learning. Pattern Recogn 125:108543
- Hausknecht M, Stone P (2015) Deep recurrent q-learning for partially observable mdps. In: 2015 Aaai fall symposium series
- Wang Y, He H, Tan X (2020) Truly proximal policy optimization. In: Uncertainty in artificial intelligence, PMLR pp 113–122
- Haarnoja T, Zhou A, Abbeel P, Levine S (2018) Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor, PMLR 1861–1870
- Edwards RD, Magee J, Bassetti WC (2018) Technical Analysis of Stock Trends. CRC Press, UK
- Wang Z, Schaul T, Hessel M, Hasselt H, Lanctot M, Freitas N (2016) Dueling network architectures for deep reinforcement learning. In: International conference on machine learning, PMLR pp 1995–2003
- Van Hasselt H, Guez A, Silver D (2016) Deep reinforcement learning with double q-learning. In: Proceedings of the AAAI conference on artificial intelligence, vol 30
- Chung J (2013) Playing atari with deep reinforcement learning. Comput Ence 21:351–362
- Ziebart BD, Maas AL, Bagnell JA, Dey AK et al (2008) Maximum entropy inverse reinforcement learning. Aaai, vol 8. IL, USA, Chicago, pp 1433–1438
- Hadfield-Menell D, Russell SJ, Abbeel P, Dragan A (2016) Cooperative inverse reinforcement learning. Advances in Neural Information Processing Systems 29

26. Herman M, Fischer V, Gindele T, Burgard W (2015) Inverse reinforcement learning of behavioral models for online-adapting navigation strategies. In: 2015 IEEE international conference on robotics and automation (ICRA), IEEE pp 3215–3222
27. Zhifei S, Joo EM (2012) A review of inverse reinforcement learning theory and recent advances. In: 2012 IEEE congress on evolutionary computation, IEEE pp 1–8
28. Audiffren J, Valko M, Lazaric A, Ghavamzadeh M (2015) Maximum entropy semi-supervised inverse reinforcement learning. In: Twenty- fourth international joint conference on artificial intelligence
29. Zhifei S, Joo EM (2012) A survey of inverse reinforcement learning techniques. *International Journal of Intelligent Computing and Cybernetics*
30. Mnih V, Badia AP, Mirza M, Graves A, Lillicrap T, Harley T, Silver D, Kavukcuoglu K (2016) Asynchronous methods for deep reinforcement learning. In: International conference on machine learning, PMLR pp 1928–1937
31. Wu Y, Mansimov E, Grosse RB, Liao S, Ba J (2017) Scalable trustregion method for deep reinforcement learning using kronecker-factored approximation. *Advances in Neural Information Processing Systems* 30
32. Ho J, Ermon S (2016) Generative adversarial imitation learning. *Advances in Neural Information Processing Systems* 29
33. Yang SY, Qiao Q, Beling PA, Scherer WT, Kirilenko AA (2015) Gaussian process-based algorithmic trading strategy identification. *Quantitative Finance* 15(10):1683–1703
34. Baiynd A-M (2011) *The trading book: a complete solution to mastering technical systems and trading psychology*. McGraw Hill Professional, NewYork
35. Liu Y, Liu Q, Zhao H, Pan Z, Liu C (2020) Adaptive quantitative trading: An imitative deep reinforcement learning approach. *Proceedings of the AAAI Conference on Artificial Intelligence* 34:2128–2135
36. Finn C, Levine S, Abbeel P (2016) Guided cost learning: Deep inverse optimal control via policy optimization. In: International conference on machine learning, PMLR pp 49–58
37. Nielsen A (2019) *Practical Time Series Analysis: Prediction with Statistics and Machine Learning*. O’ Reilly Media, London
38. Pascanu R, Çaglar G, Cho K, Bengio Y (2014) How to construct deep recurrent neural networks. *CoRR* [arXiv:1312.6026](https://arxiv.org/abs/1312.6026)
39. Li Y, Zheng W, Zheng Z (2019) Deep robust reinforcement learning for practical algorithmic trading. *IEEE Access* 7:108014–108022
40. Abbeel P, Ng AY (2004) Apprenticeship learning via inverse reinforcement learning. In: *Proceedings of the twenty-first international conference on machine learning*, p 1

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